# Lung Histopathology Triage

An AI-assisted triage platform for lung cancer screening in capacity-constrained health systems.

**Working prototype today.** A 19 MB classifier that runs entirely in a browser on commodity hardware, with no GPU and no server. Built, measured, and honest about where it fails — this document states both.

| **Seed round** | **Scale round** | **Non-dilutive** | **Model size** |
|---|---|---|---|
| $1M | $5M | Cloud + HPC credits | 19 MB, in-browser |

# The problem is throughput, not capability

Pathology is the bottleneck in every cancer pathway, and it is a people problem.

- **Demand is rising faster than the workforce.** Screening programmes and ageing populations expand the caseload; pathologist numbers are not scaling with it.
- **Triage is manual and serial.** Every slide waits for scarce specialist attention, including the large majority that are ultimately benign.
- **The cost of delay is clinical.** Time-to-diagnosis is the variable that changes outcomes, and it is governed by queue position, not by disease biology.
- **Emerging systems are hit hardest.** Laboratories in lower-resource settings carry the same caseload with a fraction of the specialist capacity — and are the least served by software priced for well-resourced hospitals.

The opportunity is not a better diagnosis. It is a cheaper, faster way to route cases so that specialist time lands where it changes an outcome.

# What already works

Not a concept. A running system, measured, with the failures documented in the next slide.

- **End-to-end pipeline.** Slide upload, preprocessing, classification, and explainability overlay, in a single-page clinical interface.
- **Runs in the browser.** The model was exported to ONNX and verified to run client-side — no GPU, no inference server, no per-request cost.
- **Numerical parity proven.** Browser inference matches the reference PyTorch implementation to **3.8e-06** on identical input. Not an approximation.
- **Deployment surface is trivial.** Static assets. It can be hosted for effectively zero marginal cost and demonstrated from a single URL.
- **Triage layer scaffolded.** The interface and pipeline that route a case after classification are in place.

# Where we are, measured

Evaluated on 120 labelled LC25000 images (40 per class). The number below is the real one, including the part that is zero.

| 66.7% | 120/120 | 0/80 | 19 MB |
|---|---|---|---|
| 3-class accuracy | benign vs malignant | adeno vs squamous | model, in-browser |

**Reading it plainly.** The model reliably separates **benign from malignant** — every one of 120 images. It does **not** separate adenocarcinoma from squamous cell carcinoma: all 80 malignant cases collapse onto a single output. It is, today, a binary benign/malignant classifier wearing a three-class head.

**And the caveat that matters more than the number.** These 120 images come from the same public dataset the model was trained on. This is **not** a held-out test set, so 100% benign/malignant recall is consistent with memorisation rather than generalisation. We report it as such rather than presenting it as clinical accuracy.

# The shortcomings, stated first

Each one is a workstream, and each workstream is what the funding buys.

- **1. Cancer subtyping does not work.** Adenocarcinoma and squamous carcinoma are indistinguishable to the current model (0/80). Subtype determines treatment, so this is the single largest functional gap.
- **2. No held-out validation.** No train/test split exists against unseen data. Without it there is no defensible accuracy claim — and no regulatory path.
- **3. Trained on public research data, not clinical data.** Real-world slides differ in staining, scanner, and preparation. The model is out-of-distribution on them, which we have observed directly.
- **4. No clinical data access yet.** Live data requires a data-sharing agreement, a data protection impact assessment, and research ethics approval. Legal review alone is not sufficient.
- **5. Explainability is approximated.** Browser inference has no gradient pass, so we use occlusion sensitivity instead of Grad-CAM. Defensible, but it needs validation against the reference method.
- **6. No regulatory approval.** The system is explicitly labelled “not for clinical use”. That is a deliberate and honest control, and lifting it is a funded programme, not a code change.

# Every gap maps to a funded workstream

This is a specificity test. Each row states what is broken, what fixes it, and what a funder can verify at the end.

| Gap | Funded workstream | Verifiable outcome |
|---|---|---|
| Subtype collapse | Architecture and loss redesign; compute-intensive retraining | Three-class accuracy on held-out data, published |
| No held-out validation | Curated splits, external validation set, evaluation harness | Reproducible accuracy + confusion matrix, third-party reviewable |
| Public data only | Clinical data partnerships; labelling and QA pipeline | Signed agreements; a validated dataset of record |
| Governance unaddressed | DPIA, ethics submission, data-sharing framework | Approved governance pack, reusable per jurisdiction |
| Explainability unvalidated | Method comparison study against Grad-CAM reference | Agreement metrics; clinician usability study |
| No regulatory path | Quality management system; regulatory strategy | Audited QMS; submission-ready technical file |

# Why now, and why this is cheap to fund

The expensive work is already done with no capital. What remains is compute and validation.

- **The hard engineering risk is retired.** The pipeline exists, runs on commodity hardware, and its numerical correctness is verified. There is no technical unknown standing between this and a validation programme.
- **Capital intensity is unusually low.** A 19 MB model that runs in a browser needs no GPU fleet and no inference infrastructure. Marginal cost per user approaches zero — which is precisely what makes deployment viable in lower-resource health systems.
- **The remaining cost is compute and rigour, not invention.** Training to separate subtypes, validating on held-out and clinical data, and building the governance and quality systems. All are well-understood, budgetable work.
- **Compute access is the lever.** Donated cloud credits and HPC allocation convert directly into validation throughput for this project — the highest-return form of non-dilutive support available.
- **Timing.** Every health system is under the same throughput pressure simultaneously. The window for a low-cost triage layer is now, not in five years.

# Use of funds

Staged so that a funder can stop after Phase 1 and still see a complete, verified result. Figures are planning targets, not quotations.

| Phase | Ask | Focus | Milestone at exit |
|---|---|---|---|
| **Phase 1** | **$1M** | Subtyping model; held-out validation; governance pack; cloud/HPC compute | Published three-class accuracy on held-out and clinical data |
| **Phase 2** | **$5M** | Clinical pilots in capacity-constrained systems; regulatory submission; team and support | Regulatory filing; validated deployment in a live pathway |
| **In kind** | Credits + HPC | Training compute, validation sweeps, evaluation infrastructure | Documented compute-to-result trail |

# The ask

Three routes in, because the support we need is not only money.

- **Sovereign and development funds — $1M seed, then $5M scale.** Health system capacity with a measurable output: time-to-diagnosis in a screening pathway. The deliverable is a validated, deployable triage layer, not a paper.
- **Cloud and compute credit programmes.** The single fastest way to accelerate this project. Credits convert straight into validation throughput, they are non-dilutive, and they carry no execution risk for the provider.
- **Supercomputing allocation.** Retraining for subtype discrimination is compute-bound. An allocation turns a twelve-month model roadmap into a quarterly one.
- **What we are not asking for.** We are not asking anyone to fund a hypothesis about a technology that might work. The prototype runs, its correctness is verified, and its weaknesses are quantified in this document. This is execution capital against a stated gap list.

**The case in one line:** a low-cost triage layer that runs on hardware already present in the clinic, with a funded plan to close the four gaps that currently stop it being clinical.

# What we will be judged on

Stated now so that progress is falsifiable. These are the milestones we hold ourselves to, in order.

- **Held-out accuracy, published.** A reproducible three-class number on data the model has never seen, with the confusion matrix released alongside it.
- **Subtype discrimination demonstrated.** Adenocarcinoma separated from squamous carcinoma at a level that is clinically useful, not merely above chance.
- **Validation on clinical data.** Performance on real slides from partner centres, with governance approved and documented.
- **Explainability validated.** The occlusion method benchmarked against the reference, and reviewed by pathologists for whether the highlighted regions are the ones a human would use.
- **A regulatory pathway opened.** Quality management system in place and a submission strategy agreed, so that lifting the “not for clinical use” label becomes a schedule rather than an aspiration.

Until every one of those is met, the system stays labelled as it is today: **retrospective validation only — not for clinical use.** We would rather be funded for closing a stated gap list than for a claim we cannot reproduce.